BASE CHANGE FOR AN OPEN IMMERSION OF SCHEMES OF FINITE TYPE OVER A FIELD, FOR TORSION COEFFICIENTS OF INVERTIBLE ORDER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the closing paragraph in which the same computation is recorded in the form used elsewhere; every reduction is written out inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree. Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invocation inside the counted span, so no load-bearing mathematics is replaced by a stub.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217). The Xy-pic square of the counted statement is delivered as written.

Allowed standard prerequisites (editorial).

- Base change map q^{-1}Rj_*F -> Rh_*p^{-1}F for a cartesian square of schemes. Same chapter, the construction of the base change map for the square displayed in the statement. Supplies the map whose being an isomorphism is asserted.
- Stalks of an etale sheaf at geometric points and vanishing of cohomology of torsion sheaves on affine curves. Same chapter, the description of stalks by cohomology over strictly henselian local rings and the vanishing results for torsion coefficients of order invertible in the base field. Supplies the two reductions, over the generic point of X and over the generic point of Y, where the transcendence degree is positive.
- Induction on the sum of the dimensions of X and Y. Same chapter, the dimension induction used in the Kunneth and base change statements of this section, together with the behaviour of the statement under restriction to open subschemes. Supplies the minimal-counterexample argument, which reduces the general case to a fibre product over a smaller base.

Source reference. Original lemma 0EZT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16410–16410. Navigation pointer invoked at proof

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line 20423; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F09: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 17975–17975. Navigation pointer invoked at proof line 20449; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19692–19692. Navigation pointer invoked at proof line 20459; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F1C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18032–18032. Navigation pointer invoked at proof line 20520; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0E46: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 20529–20529. Navigation pointer invoked at proof line 20529; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F1G: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 20370–20370. Navigation pointer invoked at proof line 20567; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F14: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 20267–20267. Navigation pointer invoked at proof line 20572; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let K be a field. Let $j : U \to X$ be an open immersion of schemes of finite type over K. Let Y be a scheme of finite type over K. Consider the diagram*

$$\begin{array}{ccc} Y \times_{\mathrm{Spec}(K)} X & \xleftarrow[h]{} & Y \times_{\mathrm{Spec}(K)} U \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xleftarrow{\quad j \quad} & U \end{array}$$

Then the base change map $q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F}$ is an isomorphism for $\mathcal{F}$ an abelian sheaf on $U_{étale}$ whose stalks are torsion of orders invertible in K.

Proof. Write $\mathcal{F} = \mathrm{colim}\, \mathcal{F}[n]$ where the colimit is over the multiplicative system of integers invertible in K. Since cohomology commutes with filtered colimits in our situation (for a precise reference see Lemma 0EZT), it suffices to prove the lemma for $\mathcal{F}[n]$. Thus we may assume $\mathcal{F}$ is a sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules for some n invertible in K (we will use this at the very end of the proof). In the proof we use the short hand $X \times_K Y$ for the fibre product over $\mathrm{Spec}(K)$. We will prove the lemma by induction on $\dim(X) + \dim(Y)$. The lemma is trivial if $\dim(X) \leq 0$, since in this

case U is an open and closed subscheme of X. Choose a point $z \in X \times_K Y$. We will show the stalk at $\overline{z}$ is an isomorphism.

Suppose that $z \mapsto x \in X$ and assume $\mathrm{trdeg}_K(\kappa(x)) > 0$. Set $X' = \mathrm{Spec}(\mathcal{O}_{X,x}^{sh})$ and denote $U' \subset X'$ the inverse image of U. Consider the base change

$$\begin{array}{ccc} Y \times_K X' & \xleftarrow{h'} & Y \times_K U' \\ \downarrow{\scriptstyle q'} & & \downarrow{\scriptstyle p'} \\ X' & \xleftarrow{j'} & U' \end{array}$$

of our diagram by $X' \to X$. Observe that $X' \to X$ is a filtered colimit of étale morphisms. By smooth base change in the form of Lemma 0F09 the pullback of $q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F}$ to X' to $Y \times_K X'$ is the map $(q')^{-1}Rj'_*\mathcal{F}' \to Rj'_*(p')^{-1}\mathcal{F}'$ where $\mathcal{F}'$ is the pullback of $\mathcal{F}$ to U'. (In this step it would suffice to use étale base change which is an essentially trivial result.) So it suffices to show that $(q')^{-1}Rj'_*\mathcal{F}' \to Rj'_*(p')^{-1}\mathcal{F}'$ is an isomorphism in order to prove that our original map is an isomorphism on stalks at $\overline{z}$. By Lemma 0F0U there is a separably closed field L/K such that $X' = \lim X_i$ with X_i affine of finite type over L and $\dim(X_i) < \dim(X)$. For i large enough there exists an open $U_i \subset X_i$ restricting to U' in X'. We may apply the induction hypothesis to the diagram

$$\begin{array}{ccc} Y \times_K X_i & \xleftarrow{h_i} & Y \times_K U_i \\ \downarrow{\scriptstyle q_i} & & \downarrow{\scriptstyle p_i} \\ X_i & \xleftarrow{j_i} & U_i \end{array} \quad \text{equal to} \quad \begin{array}{ccc} Y_L \times_L X_i & \xleftarrow{h_i} & Y_L \times_L U_i \\ \downarrow{\scriptstyle q_i} & & \downarrow{\scriptstyle p_i} \\ X_i & \xleftarrow{j_i} & U_i \end{array}$$

over the field L and the pullback of $\mathcal{F}$ to these diagrams. By Lemma 0EZT we conclude that the map $(q')^{-1}Rj'_*\mathcal{F}' \to Rj'_*(p')^{-1}\mathcal{F}$ is an isomorphism.

Suppose that $z \mapsto y \in Y$ and assume $\mathrm{trdeg}_K(\kappa(y)) > 0$. Let $Y' = \mathrm{Spec}(\mathcal{O}_{X,x}^{sh})$. By Lemma 0F0U there is a separably closed field L/K such that $Y' = \lim Y_i$ with Y_i affine of finite type over L and $\dim(Y_i) < \dim(Y)$. In particular Y' is a scheme over L. Denote with a subscript L the base change from schemes over K to schemes over L. Consider the commutative diagrams

$$\begin{array}{ccc} Y' \times_K X & \xleftarrow{h'} & Y' \times_K U \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ Y \times_K X & \xleftarrow{h} & Y \times_K U \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle p} \\ X & \xleftarrow{j} & U \end{array} \quad \text{and} \quad \begin{array}{ccc} Y' \times_L X_L & \xleftarrow{h'} & Y' \times_L U_L \\ \downarrow{\scriptstyle q'} & & \downarrow{\scriptstyle p'} \\ X_L & \xleftarrow{j_L} & U_L \\ \downarrow & & \downarrow \\ X & \xleftarrow{j} & U \end{array}$$

and observe the top and bottom rows are the same on the left and the right. By smooth base change we see that $f^{-1}Rh_*p^{-1}\mathcal{F} = Rh'_*(f')^{-1}p^{-1}\mathcal{F}$ (similarly to the previous paragraph). By smooth base change for $\mathrm{Spec}(L) \to \mathrm{Spec}(K)$ (Lemma 0F1C) we see that $Rj_{L,*}\mathcal{F}_L$ is the pullback of $Rj_*\mathcal{F}$ to X_L. Combining these two observations, we conclude that it suffices to prove the base change map for the upper square in the diagram on the right is an isomorphism in order to prove that our original map is an isomorphism on stalks at $\overline{z}$[1]. Then using that $Y' = \lim Y_i$ and argueing exactly as in the previous paragraph we see that the induction hypothesis forces our map over $Y' \times_K X$ to be an isomorphism.

[1]Here we use that a "vertical composition" of base change maps is a base change map as explained in Cohomology on Sites, Remark 0E46.

Thus any counter example with $\dim(X) + \dim(Y)$ minimal would only have nonisomorphisms $q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F}$ on stalks at closed points of $X \times_K Y$ (because a point z of $X \times_K Y$ is a closed point if and only if both the image of z in X and in Y are closed). Since it is enough to prove the isomorphism locally, we may assume X and Y are affine. However, then we can choose an open dense immersion $Y \to Y'$ with Y' projective. (Choose a closed immersion $Y \to \mathbf{A}^n_K$ and let Y' be the scheme theoretic closure of Y in $\mathbf{P}^n_K$.) Then $\dim(Y') = \dim(Y)$ and hence we get a "minimal" counter example with Y projective over K. In the next paragraph we show that this can't happen.

Consider a diagram as in the statement of the lemma such that $q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F}$ is an isomorphism at all non-closed points of $X \times_K Y$ and such that Y is projective. The restriction of the map to $(X \times_K Y)_{K^{sep}}$ is the corresponding map for the diagram of the lemma base changed to K^{sep}. Thus we may and do assume K is separably algebraically closed. Choose a distinguished triangle

$$q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F} \to Q \to (q^{-1}Rj_*\mathcal{F})[1]$$

in $D((X \times_K Y)_{étale})$. Since Q is supported in closed points we see that it suffices to prove $H^i(X \times_K Y, Q) = 0$ for all i, see Lemma 0F1G. Thus it suffices to prove that $q^{-1}Rj_*\mathcal{F} \to Rh_*p^{-1}\mathcal{F}$ induces an isomorphism on cohomology. Recall that $\mathcal{F}$ is annihilated by n invertible in K. By the Künneth formula of Lemma 0F14 we have

$$\begin{aligned} R\Gamma(X \times_K Y, q^{-1}Rj_*\mathcal{F}) &= R\Gamma(X, Rj_*\mathcal{F}) \otimes^{\mathbf{L}}_{\mathbf{Z}/n\mathbf{Z}} R\Gamma(Y, \mathbf{Z}/n\mathbf{Z}) \\ &= R\Gamma(U, \mathcal{F}) \otimes^{\mathbf{L}}_{\mathbf{Z}/n\mathbf{Z}} R\Gamma(Y, \mathbf{Z}/n\mathbf{Z}) \end{aligned}$$

and

$$R\Gamma(X \times_K Y, Rh_*p^{-1}\mathcal{F}) = R\Gamma(U \times_K Y, p^{-1}\mathcal{F}) = R\Gamma(U, \mathcal{F}) \otimes^{\mathbf{L}}_{\mathbf{Z}/n\mathbf{Z}} R\Gamma(Y, \mathbf{Z}/n\mathbf{Z})$$

This finishes the proof. □